Policy Gradient in RL

Dr. Dibyajyoti Guha

Assistant Professor

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Outline

- Policy Gradient Derivation
- Temporal Decomposition
- Baseline Subtraction
- Value Function Estimation

Policy Optimization

Consider control policy parameterized by parameter vector θ.
$\arg\max_\theta \mathbb{E}\Big[\sum_{t=0}^{H} R(s_t)|\pi(\theta)\Big]$
Stochastic policy smooth out the problem
$\pi_\theta(u|s)$: probability of action u in state s.
Let τ denote a state-action sequence $s_0, u_0, \cdots, s_H, u_H$. We overload notation: $R(\tau) = \sum_{t=0}^{H} R(s_t, u_t)$.

$$U(\theta) = \mathbb{E}\Big[\sum_{t=0}^{H} R(s_t, u_t); \pi(\theta)\Big] = \sum_{\tau} P(\tau;\theta)R(\tau) \tag{1}$$

Our goal is to find θ:

$$\max_{\theta} U(\theta) = max_\theta \sum_{\tau} P(\tau;\theta)R(\tau) \tag{2}$$

Likelihood Ratio Policy Gradient

$U(\theta) = \sum_\tau P(\tau;\theta)R(\tau)$, Taking the gradient w.r.t. θ gives

$$\begin{aligned}
\nabla_\theta U(\theta) &= \nabla_\theta \sum_\tau P(\tau;\theta)R(\tau) = \sum_\tau \nabla_\theta P(\tau;\theta)R(\tau) \\
&= \sum_\tau \frac{P(\tau;\theta)}{P(\tau;\theta)} \nabla_\theta P(\tau;\theta)R(\tau) \\
&= \sum_\tau P(\tau;\theta) \frac{\nabla_\theta P(\tau;\theta)}{P(\tau;\theta)} R(\tau) \\
&= \sum_\tau P(\tau;\theta) \nabla_\theta \log P(\tau;\theta) R(\tau) \qquad (3)
\end{aligned}$$

Approximate with the empirical estimate for m sample paths under policy $\pi(\theta)$:

$$\nabla_\theta U(\theta) \equiv \frac{1}{m} \sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta) R(\tau^{(i)}) \qquad (4)$$

Likelihood Ratio Policy Gradient

$$\nabla_\theta U(\theta) \equiv \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta) R(\tau^{(i)})$$

Gradient tries to:

- Increase probability of paths with positive R
- Decrease probability of path with negative R

Likelohood ratio changes probabilities of experienced paths; does not try to change the paths (path derivaties) .

Decompose Paths into States and Actions

$$
\begin{aligned}
\nabla_\theta \log P(\tau^{(i)};\theta) &= \nabla_\theta \log \left[\prod_{t=0}^{H} \underbrace{P(s_{t+1}^{(i)} | s_t^{(i)}, u_t^{(i)})}_{\text{dynamics model}} \cdot \underbrace{\pi_\theta(u_t^{(i)} | s_t^{(i)})}_{\text{policy}} \right] \\
&= \nabla_\theta \left[\sum_{t=0}^{H} \log P(s_{t+1}^{(i)} | s_t^{(i)}, u_t^{(i)}) + \sum_{t=0}^{H} \log \pi_\theta(u_t^{(i)} | s_t^{(i)}) \right] \\
&= \nabla_\theta \sum_{t=0}^{H} \log \pi_\theta(u_t^{(i)} | s_t^{(i)}) \\
&= \sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(u_t^{(i)} | s_t^{(i)}) \qquad (5)
\end{aligned}
$$

To increase the probability of a trajectory you need to increase the log probability of the actions along the trajectory.

Gradient estimate

We had started with estimation of $\hat{g} = \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta)R(\tau^{(i)})$
After temporal decomposition we got:

$$\nabla_\theta \log P(\tau^{(i)};\theta) = \sum_{t=0}^{H} \underbrace{\nabla_\theta \log \pi_\theta(u_t^{(i)}|s_t^{(i)})}_{\text{no dynamics model needed!!!}}$$

You can roll out the current policy, you know the rewards along each trajectory, state-action pairs along each trajectory

Baseline Subtraction

We had started with estimation of
$\nabla_\theta U(\theta) \equiv \hat{g} = \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta) R(\tau^{(i)})$
Consider subtracting baseline:
$\nabla_\theta U(\theta) \equiv \hat{g} = \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta)(R(\tau^{(i)}) - b)$. Its OK as long as baseline doesn't depend on action in logprob(action)

$$\mathbb{E}[\nabla_\theta \log P(\tau;\theta) b] = \sum_\tau P(\tau;\theta)\nabla_\theta \log P(\tau;\theta) b$$

$$= \sum_\tau P(\tau;\theta)\frac{\nabla_\theta P(\tau;\theta)}{P(\tau;\theta)} b$$

$$= \sum_\tau \nabla_\theta P(\tau;\theta) b$$

$$= \nabla_\theta \sum_\tau P(\tau;\theta) b$$

$$= b\nabla_\theta(\sum_\tau P(\tau;\theta)) = b \times 0$$

More Temporal Structure and Baseline

$$\hat{g} = \frac{1}{m}\sum_{i=1}^{m} \nabla_\theta \log P(\tau^{(i)};\theta)(R(\tau^{(i)}) - b)$$

$$= \frac{1}{m}\sum_{i=1}^{m} \left(\sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(u_t^{(i)}|s_t^{(i)})\right)\left(\sum_{t=0}^{H} R(s_t^{(i)}, u_t^{(i)}) - b\right)$$

$$= \frac{1}{m}\sum_{i=1}^{m} \left(\sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(u_t^{(i)}|s_t^{(i)})\right)\left[\underbrace{\sum_{k=0}^{t-1} R(s_k^{(i)}, u_k^{(i)})}_{\text{not dependent on } u_t^{(i)}} + \sum_{k=t}^{H} R(s_k^{(i)}, u_k^{(i)}) - b\right]$$

Removing terms that dont depend on current action can lower variance

$$= \frac{1}{m}\sum_{i=1}^{m}\sum_{t=0}^{H} \nabla_\theta \log \pi_\theta(u_t^{(i)}|s_t^{(i)})\left(\sum_{k=t}^{H} R(s_k^{(i)}, u_k^{(i)}) - b(s_t^{(i)})\right) \tag{6}$$

Why would change the probability of an action at time step t based on the reward you received from $i = o \cdots t-1$ time steps?

- Constant baseline: $b = \mathbb{E}[R(\tau)] \equiv \frac{1}{m}\sum_{i=1}^{m} R(\tau^{(i})$
- time dependent baseline: $b_t = \frac{1}{m}\sum_{i=1}^{m}\sum_{k=t}^{H} R(s_k^{(i)}, u_k^{(i)})$
- State dependent expected return:
  $b(s_t) = \mathbb{E}[r_t + r_{t+1} + r_{t+2} + \cdots + r_H] = V^{\pi}(s_t)$

Increase log probability of action proportionally to how much its returns are better than the expected return under the current policy

Value Function Estimation: Monte Carlo estimation of V^{π}

$$\frac{1}{m}\sum_{i=1}^{m}\sum_{t=0}^{H-1}\nabla_\theta \log \pi_\theta(u_t^{(i)}|s_t^{(i)})\Big(\sum_{k=t}^{H-1} R(s_k^{(i)}, u_k^{(i)}) - \underbrace{V^{\pi}(s_k^{(i)})}_{\text{How to estimate?}}\Big)$$

Init $V^{\pi}_{\phi_0}$
Collect trajectories $\tau_1, \tau_2, \cdots, \tau_m$
Regress against empirical return:

$$\phi_{i+1} \leftarrow \arg\min_{\phi} \frac{1}{m}\sum_{i=1}^{m}\sum_{t=0}^{H-1}\Big(V^{\pi}_{\phi}(s_t^{(i)}) - \sum_{k=t}^{H-1} R(s_k^{(i)}, u_k^{(i)})\Big)^2 \tag{7}$$

Importance Sampling

$$U(\theta) = \mathbb{E}_{\tau \sim \theta_{old}} \left[\frac{P(\tau|\theta)}{P(\tau|\theta_{old})} R(\tau) \right]$$

$$\nabla_\theta U(\theta) = \mathbb{E}_{\tau \sim \theta_{old}} \left[\frac{\nabla_\theta P(\tau|\theta)}{P(\tau|\theta_{old})} R(\tau) \right]$$

$$\nabla_\theta U(\theta)|_{\theta=\theta_{old}} = \mathbb{E}_{\tau \sim \theta_{old}} \left[\frac{\nabla_\theta P(\tau|\theta)|_{\theta=\theta_{old}}}{P(\tau|\theta_{old})} R(\tau) \right]$$

$$= \mathbb{E}_{\tau \sim \theta_{old}} \left[\nabla_\theta \log P(\tau|\theta)|_{\theta_{old}} R(\tau) \right]$$

If you do an update, then you want to measure how good was the update? This equation will help in that.
We can use importance sampled objective as "surrogate loss"

IS in Policy Gradient

$$\nabla_\theta U(\theta) \equiv \mathbb{E}_{(s_t,a_t)\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) A(s_t,a_t)\right]$$

$$= \mathbb{E}_{(s_t,a_t)\sim\pi_{\theta_{old}}}\left[\frac{\pi_\theta(s_t,a_t)}{\pi_{\theta_{old}}(s_t,a_t)} \nabla_\theta \log \pi_\theta(a_t|s_t) A(s_t,a_t)\right] \tag{8}$$

Surrogate objective function

$$J(\theta) = \mathbb{E}_{(s_t,a_t)\sim\pi_{\theta_{old}}}\left[\frac{\pi_\theta(s_t,a_t)}{\pi_{\theta_{old}}(s_t,a_t)} A(s_t,a_t)\right] \tag{9}$$

Variance in Policy Gradient

$$\mathbb{E}_{x\sim p}[f(x)] = \mathbb{E}_{x\sim q}[f(x)\frac{p(x)}{q(x)}]$$
$$Var_{x\sim p}[f(x)] = \mathbb{E}_{x\sim p}[(f(x))^2] - (\mathbb{E}_{x\sim p}[f(x)])^2 \tag{10}$$